

Clostridioides difficile infections and antimicrobial consumption: a double-edged sword

The association between Clostridioides difficile infections and antimicrobial consumption in Belgian hospitals between 2014 and 2023

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BACKGROUND

Clostridioides difficile infections (CDIs) have a high clinical, economic, and social burden. In European hospitals, CDIs are responsible for 5.9% of all healthcare-associated infections. On one hand, risk factors, such as antibiotic therapy, can cause dysbiosis of the gut microbiome, leading to a CDI. On the other hand, antibiotics are used for the treatment of CDI.

This study examined the association between hospital-associated (HA)-CDI and antimicrobial consumption (AMC) in Belgian hospitals from 2014 to 2023.

Antibiotics are a risk factor for the development of CDI

e.g. Beta-lactam/beta-lactamase inhibitors, cephalosporins (2nd, 3rd, and 4th generation), carbapenems, clindamycin, fluoroquinolones, and glycopeptides.

Antibiotics are a treatment for CDI

e.g. metronidazole, vancomycin, fidaxomicin.

METHODOLOGY

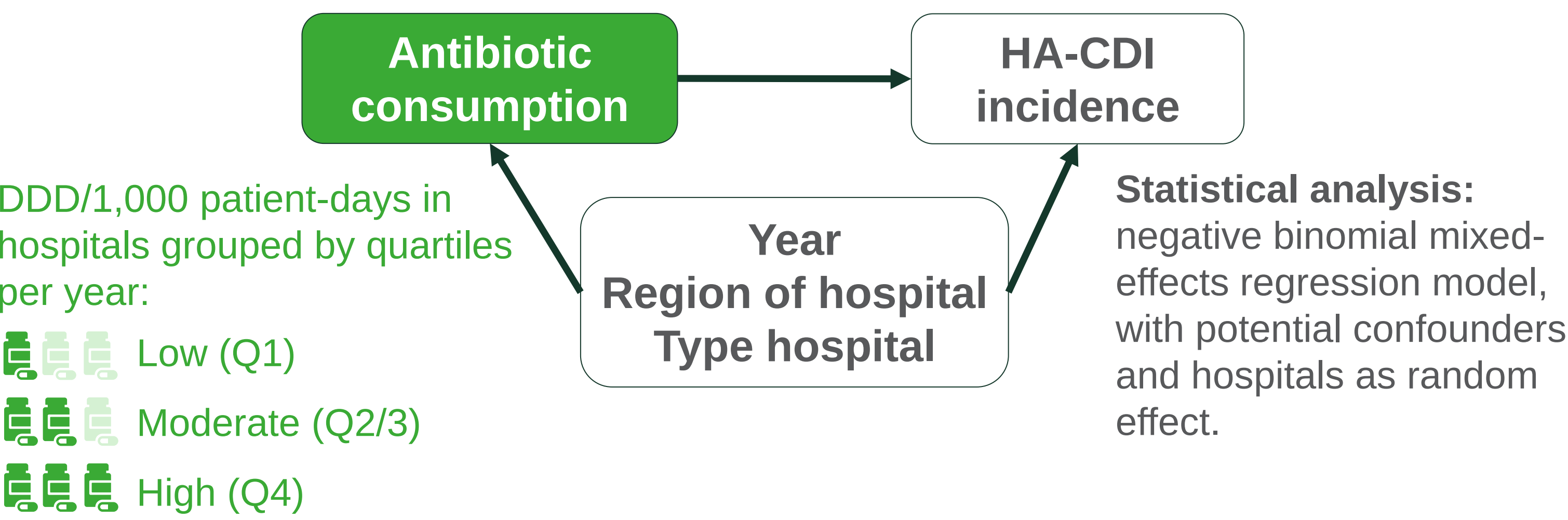
Retrospective observational study with **two surveillance datasets** from hospitals between 2014 and 2023

Surveillance CDI

- Reported by hospitals
- Partly mandatory** participation
- At least one semester per year
- Patient-level** data
- HA-CDI per 10,000 patient-days**: CDI that occur after two or more days of hospital admission

Surveillance AMC

- Through reimbursement data
- Complete coverage**
- Collected per quarter
- Hospital-level** data
- Antibiotic selection by Anatomical Therapeutic Chemical code
- Defined Daily Doses (DDD) per 1,000 patient-days**



RESULTS & DISCUSSION

- On average, **82% of hospitals (87/106)** participated
- Only statistically significant association found for: **Carbapenems** and **glycopeptides** (Table 1)
- Of note: Glycopeptides contain vancomycin, which is a treatment for CDI
- Region and type of hospital had an independent effect on HA-CDI incidence
- Secondary hospitals exhibited 37% higher HA-CDI incidence rates compared to primary hospitals (p-value = 0.007), even with region and antimicrobial consumption in the model



Table 1: Incidence rate ratios (IRR) with 95% confidence intervals (95% CI) and p-values of the association between consumption (DDD per 1,000) of antibiotics considered as risk factors for the development of *Clostridioides difficile* infections and incidence of hospital-associated *Clostridioides difficile* infections. Belgium, 2014-2023.

Predictors		Beta lactam/beta lactamase inhibitors		2nd, 3rd and 4th generation cephalosporins		Carbapenems	
		IRR (95% CI)	p-value	IRR (95% CI)	p-value	IRR (95% CI)	p-value
Consumption group	Low	ref	ref	ref	ref	ref	ref
	Moderate	1.03 (0.90-1.18)	0.675	1.11 (0.97-1.27)	0.114	1.24 (1.10-1.40)	0.001
	High	0.98 (0.83-1.16)	0.783	1.13 (0.94-1.37)	0.202	1.32 (1.13-1.55)	<0.001
Predictors		Clindamycin		Fluoroquinolones		Glycopeptides	
		IRR (95% CI)	p-value	IRR (95% CI)	p-value	IRR (95% CI)	p-value
Consumption group	Low	ref	ref	ref	ref	ref	ref
	Moderate	1.05 (0.94-1.18)	0.372	1.12 (0.99-1.26)	0.065	1.32 (1.17-1.49)	<0.001
	High	1.08 (0.92-1.26)	0.365	1.16 (1.00-1.35)	0.055	1.34 (1.14-1.57)	<0.001

CONCLUSION

- These findings suggest that while some antibiotics may be a key driver of HA-CDI, other facility-related factors likely contribute.
- Strengthening antimicrobial stewardship and infection prevention and control measures is essential to reduce HA-CDI and antimicrobial resistance.

REFERENCES

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